Web Application Fundamentals

- How does a browser communicate with a web server? Through HTTP or HTTPS requests and responses.

- What is the difference between HTTP and HTTPS? HTTPS encrypts communication using TLS, while HTTP does not.

- Why are sessions required in web applications? To maintain user state and recognize users across multiple requests.

Web Vulnerabilities

- What causes SQL Injection? Improperly handling user input in SQL queries.

- How does XSS affect users? It injects malicious scripts into web pages, potentially stealing data or hijacking sessions.

- Why does CSRF depend on user sessions? Because the browser automatically sends session cookies with requests, potentially allowing unauthorized actions.

- Why are file uploads security-sensitive? Malicious files may contain executable code or exploit server vulnerabilities.

- What makes command injection dangerous? It allows attackers to execute unauthorized system commands.

- Why is authentication security important? To prevent unauthorized access to accounts and sensitive information.

Metasploit

- What is the purpose of the Metasploit Framework? To test vulnerabilities and validate security controls through authorized security testing.

- Why must security testing be authorized? To ensure legal compliance and prevent harm to systems and data.

- Why are reporting and documentation important? To record findings, assess risks, and recommend security improvements.